

Database Technologies - Assignment 2

Write the `SELECT` queries to do the following:-

Note : To solve below queries use "sales" database

1. Write a query that produces all rows from the Customers table for which the salesperson's number is 1001.

```
mysql> SELECT * FROM customers
-> WHERE snum = 1001;
+-----+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+-----+
| 2001 | Hoffman | London | 100    | 1001 |
| 2006 | Clemens | London | 100    | 1001 |
+-----+-----+-----+-----+-----+
2 rows in set (0.048 sec)
```

2. Write a select command that produces the rating followed by the name of each customer in San Jose.

```
mysql> SELECT rating, cname FROM customers
-> WHERE city='San Jose';
+-----+-----+
| rating | cname   |
+-----+-----+
| 200    | Liu     |
| 300    | Cisneros |
+-----+-----+
2 rows in set (0.040 sec)
```

3. Write a query that will produce the snum values of all salespeople from the Orders table (with the duplicate values suppressed).

```
mysql> SELECT DISTINCT snum FROM orders;
+-----+
| snum |
+-----+
| 1007 |
| 1001 |
| 1004 |
| 1002 |
| 1003 |
+-----+
5 rows in set (0.036 sec)
```

4. Write a query that will display all the orders for amount more than Rs. 1,000.

```
mysql> SELECT * FROM orders
-> WHERE amt>1000;
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
6 rows in set (0.035 sec)
```

5. Write a query that will give you the names and cities of all salespeople in London with a commission above 0.10.

```
mysql> SELECT sname, city FROM salespeople
-> WHERE city='London' AND comm>0.10;
+-----+-----+
| sname | city |
+-----+-----+
| Peel  | London |
| Motika | London |
+-----+-----+
2 rows in set (0.010 sec)
```

6. Write an SQL query that returns all customers who have a rating greater than 100, along with the customers located in Rome regardless of their rating.

```
mysql> SELECT * FROM customers
-> WHERE rating>100 OR city='Rome';
+-----+-----+-----+-----+-----+
| cnum | cname   | city    | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome    | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin  | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
| 2007 | Pereira  | Rome    | 100    | 1004 |
+-----+-----+-----+-----+-----+
5 rows in set (0.062 sec)
```

7. What will be the output from the following query?

Select * from Orders where (amt < 1000 OR NOT (odate = '1990-10-03' AND cnum > 2003));

```
mysql> SELECT * FROM orders;
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

10 rows in set (0.032 sec)

8. What will be the output of the following query?

Select * from Orders where NOT (odate = '1990-10-03' OR snum > 1006) AND amt >= 1500;

```
mysql> SELECT * FROM orders;
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

10 rows in set (0.032 sec)

```
mysql> Select * from Orders where NOT (odate = '1990-10-03' OR snum >1006) AND amt >
= 1500;
+-----+-----+-----+-----+-----+
| onum | amt      | odate      | cnum | snum |
+-----+-----+-----+-----+-----+
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.051 sec)
```

9. What is a simpler way to write this query?
 Select snum, sname, city, comm from Salespeople
 Where (comm >= .12 AND comm <= .14);

```
mysql> SELECT * FROM salespeople
-> WHERE comm BETWEEN 0.12 AND 0.14;
+-----+-----+-----+-----+
| snum | sname   | city      | comm |
+-----+-----+-----+-----+
| 1001 | Peel    | London    | 0.12 |
| 1002 | Serres  | San Jose  | 0.13 |
+-----+-----+-----+-----+
2 rows in set (0.049 sec)
```

10. Write a query that selects all orders except those
 with amount less than 100.

```
mysql> SELECT * FROM orders
-> WHERE amt>=100;
+-----+-----+-----+-----+-----+
| onum | amt      | odate      | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19   | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10  | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45  | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16  | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23  | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00  | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95   | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88  | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
8 rows in set (0.053 sec)
```